PROPRIETARY & CONFIDENTIAL — LWANDILE ENGINEERING PROJECTS & INCLUSIVE ENTREPRENEURIAL ACADEMY SCAPUSH PRECISION STANDARD

DETAILED CAPEX LOCK-IN BREAKDOWN: ZAR 3,734,500.00

Sovereign Bio-Solar-Thermal Campus | Itemized Labor, Machinery & Material Analysis (20,000m² Fence Infused)

Doc Ref: 25_itemized_capex_lockin.html | Status: FINAL COSTING LOCK-IN (FENCED BASELINE) | Governance: Scapush Precision | South African Market Benchmark

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1. Cost-Engineering Verification & Market Rate Basis

LISTEN SECTION

This document establishes the bottom-up mathematical justification to lock in the **ZAR 3,734,500.00 Scaled MVP Budget** (inclusive of the 20,000m² perimeter fence and 10% contingency). All hourly labor tariffs align with SAFCEC (South African Federation of Civil Engineering Contractors) and bargaining council benchmarks, machinery wet-rate tariffs reflect regional plant hire averages in Mpumalanga, and equipment procurement figures represent verified Tier-1 vendor quotes.

CAPEX FINANCIAL RECONCILIATION SUMMARY

- **Labor Category Allocation (5,640 Man-Hours):** R 913,000.00
- **Machinery & Plant Hire Allocation (530 Machine-Hours):** R 492,000.00
- **Materials, Equipment & Freight Procurement (Incl. 20,000m² Fence):** R 1,990,000.00
- **Subtotal Direct CAPEX:** R 3,395,000.00
- **Mandatory 10% Contingency Buffer:** R 339,500.00
- **TOTAL LOCKED CAPEX BASELINE: ZAR 3,734,500.00**

2. Labor Force Breakdown by Hourly Market Rate (5,640 Man-Hours)

LISTEN SECTION

Skill Category / Trade Role	Total Allocated Hours	Benchmark Market Rate (ZAR / Hour)	Calculated Total Cost (ZAR)	Primary Scope Responsibility
Lead Systems Engineer / Project Manager	480 Hours	R 450.00 / hr	R 216,000.00	Overall technical integration, SCADA programming, biosecurity & JET compliance.
Certified Master Electrician / PV Wireman	520 Hours	R 280.00 / hr	R 145,600.00	30kWp PV array, 60kWh BESS, 30kVA inverter, DB panels, and 30kW EV charger.

Skill Category / Trade Role	Total Allocated Hours	Benchmark Market Rate (ZAR / Hour)	Calculated Total Cost (ZAR)	Primary Scope Responsibility
Specialized Gas Fitter / Mechanical Plumber	460 Hours	R 250.00 / hr	R 115,000.00	70m³ digester gas lines, scrubbers, 15kW CHP piping, 8-pod underfloor PEX loops.
Master Mason / Bricklaying Specialist	640 Hours	R 180.00 / hr	R 115,200.00	70m³ digester dome masonry, Barn 1 & Barn 2 walls, biosecurity station.
Structural Steel Erector / Fencing / Welder	700 Hours	R 190.00 / hr	R 133,000.00	20,000m² perimeter fence (2.2m height), barn superstructures, 6m water tower, solar frames.
Instrumentation & Controls Technician	320 Hours	R 260.00 / hr	R 83,200.00	120 SCADA sensors calibration, PLC wiring, HMI configuration, GSM dialer.
Civil & General Artisans / Paving Crew	2,520 Hours	R 45.00 / hr	R 113,000.00	Trenching, rebar tying, concrete pouring support, Zone M step-free paving.
		TOTAL LABOR FORCE ALLOCATION:	**R 913,000.00**	**5,640 Total Man-Hours**

3. Machinery & Plant Rental Breakdown by Hourly Wet-Rate (530 Hours)

LISTEN SECTION

Machinery / Heavy Equipment Type	Allocated Operating Hours	Regional Wet-Rate (ZAR / Hour)	Calculated Total Cost (ZAR)	Specific Site Utilization Scope
Hydraulic Rotary Rig (Truck-Mounted)	32 Hours	R 2,500.00 / hr	R 80,000.00	Drilling Borehole-01 to 100m depth, 165mm uPVC casing installation.
20-Ton Excavator with Breaker	64 Hours	R 1,350.00 / hr	R 86,400.00	Site pad earthworks, 70m³ digester underground bowl excavation.
4x4 Heavy Utility TLB	120 Hours	R 620.00 / hr	R 74,400.00	Foundation trenching for Barns 1 & 2, water pipe/cable backfilling, fence post holes.
25-Ton Mobile Hydraulic Crane	24 Hours	R 2,200.00 / hr	R 52,800.00	Lifting 6m water tank tower, BESS container, 15kW CHP set, 12m Academy container.
10-Ton Smooth Drum Vibratory Roller	32 Hours	R 850.00 / hr	R 27,200.00	Sub-base compaction for Barn slabs, Zone M paving, digester base.
Ready-Mix Concrete Transit Mixers (18 Trips)	48 Hours	R 2,800.00 / hr (Mix+Trip)	R 134,400.00	30 MPa concrete supply for Barn slabs (768m²), digester base, tower footings.
On-site Mixers, Rammers & Compactors	210 Hours	R 175.00 / hr	R 36,800.00	Localized mortar mixing, trench tamping, pathway edge compaction.
		TOTAL MACHINERY & PLANT ALLOCATION:	**R 492,000.00**	**530 Total Machine-Hours**

4. Material & Equipment Procurement Breakdown (Including Perimeter Fence)

LISTEN SECTION

Procurement Package	Itemized Major Hardware & Material Specifications	Calculated Cost (ZAR)
20,000m² Perimeter Security Fence	**2.2m high heavy-duty security fence (approx. 570m perimeter run, security gates & posts)**	**R 185,000.00**
Solar PV & BESS Hardware	30kWp Tier-1 PV panels, ground frames, 60kWh LFP BESS, 30kVA Hybrid Inverter, Zone L container.	R 580,000.00
Barns 1 & 2 Superstructures	Steel portal frames, IBR roofing, wall cladding, 2% sloped concrete rebar, drinker/feeder lines.	R 410,000.00
Bio-Energy & CHP Hardware	70m³ digester liner/gasholder, 15kW CHP engine set, gas scrubbers, flue heat exchangers.	R 320,000.00
Water & Hydronic Piping	1.5kW DC solar pump, 2x10kL tanks, 6m steel tower, UV plant, 20mm PEX-a loops, 50mm XPS insulation.	R 215,000.00
SCADA, EV & Academy Fit-out	PLC panel, 120 sensors, 30kW DC EV charger, 4x wheelchair bays, 12m container classroom fit-out.	R 180,000.00
Biosecurity & Sanitation	Shower container, vehicle spray arch pump/nozzles, chemical dosing unit, PPE lockers.	R 100,000.00
	TOTAL MATERIAL & EQUIPMENT ALLOCATION:	**R 1,990,000.00**

5. Master CAPEX Reconciliation to ZAR 3,734,500.00

LISTEN SECTION

Budget Component Category	Subtotal Allocation (ZAR)	Percentage of Direct CAPEX
Total Labor Force Costs (5,640 Man-Hours)	R 913,000.00	26.89 %
Total Machinery & Plant Hire Costs (530 Hours)	R 492,000.00	14.50 %
Total Materials & Equipment Procurement (Incl. Fence)	R 1,990,000.00	58.62 %
SUBTOTAL DIRECT CAPEX:	**R 3,395,000.00**	**100.00 %**
MANDATORY 10% CONTINGENCY BUFFER:	**R 339,500.00**	**10.00 %**
TOTAL LOCKED SCALED MVP CAPEX BASELINE (FENCED):	**ZAR 3,734,500.00**	**LOCKED IN**

6. Approval & Authorization

LISTEN SECTION

DOCUMENT STATUS

STATUS: APPROVED FINAL BASELINE (REV 6.0) — COSTING LOCKED AT ZAR 3,734,500.00 INCLUSIVE OF 20,000M² FENCE FOR JET EOI SUBMISSION

PROJECT STEWARD & TECHNICAL LEAD



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Operational Footprint

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